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Characteristics of energy cascade in Iceland Basin and Irminger Sea

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Abstract

The subpolar North Atlantic exhibits marked seasonal variability in mesoscale and submesoscale kinetic energy (KE), yet direct observational evidence of cross-scale KE cascades remains limited. Leveraging high-resolution velocity measurements from Shipborne Acoustic Doppler Current Profiler (SADCP) transects, this study investigates the characteristics and seasonal modulation of kinetic energy cascades and energy injection processes in the Iceland Basin and Irminger Sea. Contrary to previous model-based findings, no clear bifurcation between forward and inverse cascades is observed above the 4 km scale; instead, inverse cascades dominate throughout the region. These inverse cascades show strong seasonal variability—peaking in spring, weakening in fall, and reaching a minimum in summer. Distinct energy injection scales are identified at approximately 4–8 km and 13 km in both basins, with their intensities exhibiting parallel seasonal behavior. Importantly, these seasonal variations in inverse kinetic energy cascades and energy injections do not significantly impact the mesoscale kinetic energy patterns, indicating that mesoscale eddies in this region draw on additional, more dominant energy sources.

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1. Introduction

Oceans are governed by a variety of dynamical processes, ranging from large-scale circulations to mesoscale eddies and submesoscale motions. Mesoscale eddies, which dominate oceanic kinetic energy, contribute up to 70% of the ocean's total kinetic energy (KE) (Tulloch et al. 2011; Ferrari et al. 2009). Understanding the energy cascades of mesoscale eddies is essential for revealing the generation and dissipation of kinetic energy in the ocean. Mesoscale eddies, influenced by Earth's rotation and ocean stratification, can be described using quasi-geostrophic (QG) dynamics theory (Salmon 1978; Charney 1971; Lapeyre and Klein 2006). This framework exhibits similarities to two-dimensional turbulence, where kinetic energy undergoes an inverse cascade, transferring kinetic energy from smaller scales to larger scales (Vallis, 2017). However, only about 10% of the kinetic energy from mesoscale eddies is dissipated at larger scales through boundary friction (Wunsch et al. 2004; Deng and Wang 2024), leaving the dissipation mechanisms of mesoscale eddies as an unresolved problem. Recent studies suggests that positive kinetic energy cascades at submesoscale may play a significant role in this dissipation process (Garabato et al.2022; Poje et al. 2017; Galperin et al. 2021).

At the submesoscale, where the Rossby number approaches 1, vertical velocities become significant. This may lead to three-dimensional turbulence and forward kinetic energy cascades under certain conditions (Jing et al. 2022; Cao et al. 2021). At the submesoscale, there are also significant pathway for energy injections. Submesoscale processes, such as mixed-layer baroclinic instabilities, play a key role in converting potential energy into kinetic energy (Boccaletti et al. 2007). Combined with the inverse cascade process, the energy cascade framework is further supported by observational and modeling studies showing bi-directional cascades, where kinetic energy transfer both forward to smaller dissipative scales and inversely to mesoscale eddies (Klein et al., 2019).

While theoretical and numerical studies support the presence of bidirectional cascades, the precise transition scale remains debated. The kinetic inverse cascading structure was validated in numerical simulations by Schlösser and Eden (2007), who

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used a 1/12° eddy-resolving version of MOM2 for the North Atlantic Ocean. Their spectral analysis revealed an inverse cascade of kinetic energy near the surface and a forward cascade of potential energy, highlighting the coexistence of different energy transfer pathways in the upper ocean. In parallel, studies based on high-resolution models indicate that the forward kinetic energy cascades are on the scale of $o(1) - o(10)$ km. Submesoscale processes, particularly mixed-layer baroclinic instability and frontogenesis, play a critical role at these smaller scales (Boccaletti et al., 2007; Fox-Kemper et al., 2008, Held et al., 1995).

Observing kinetic energy cascades require high spatial resolution, making such studies particularly challenging. As a result, existing observation-based studies are few and primarily concentrated in the mid- and low-latitude regions of the North Atlantic and North Pacific (Qiu et al. 2022; Balwada et al. 2016; 2022). The observations have corroborated the bidirectional structure of the kinetic energy cascade, but the bifurcation points of the bidirectional cascade are different in each region. In the Gulf of Mexico, the bifurcation points of the bidirectional kinetic energy cascade exhibit clear seasonal variations, approaching 1 km in summer and 5 km in winter. These variations are primarily driven by mixed layer baroclinic instabilities, which result from seasonal changes in mixed layer depth (Balwada et al., 2022). In the Pacific Ocean region, work by Qiu (2022) shows that the bifurcation points of bidirectional kinetic energy cascades are latitude dependent, ranging from 200km at low latitude region to 8km at middle latitude region. However, high-latitude regions have yet to confirm these bifurcation points due to observational limitations.

In the subpolar North Atlantic, the characteristics of baroclinic instability differ significantly from those at mid- and low latitudes. In mid- and low- latitudes regions, mixed-layer baroclinic instabilities (MLIs) typically occur at scales smaller than the first baroclinic Rossby deformation radius and are linked to the bifurcation point of the bidirectional energy cascade (Balwada et al., 2022). In contrast, wintertime mixed layers in the subpolar North Atlantic can reach depths of several hundred meters. Under such conditions, combined with weak stratification, MLIs develop at larger horizontal scales set by the balance between stratification, vertical shear, and lateral buoyancy

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gradients. According to Boccaletti et al. (2007), their characteristic scale increases with decreasing Richardson number and follows: $\frac{L}{L_R} \approx \sqrt{1 + Ri^{-1}}$, where L_R denotes the local Rossby deformation radius, L denotes the mixed layer deformation radius. When $Ri \approx 1$, the instability scale can exceed L_R, especially in weakly stratified, deep mixed layers. Therefore, submesoscale instabilities in this region can occur at scales of $O(10)$km or more—potentially larger than the climatological deformation radius of ~20 km (Chelton et al., 1998).

Moreover, linear theoretical studies have found that the dominant vertical modes and horizontal scales of baroclinic instability in this region are more diverse than in mid- and low-latitude areas, indicating a more complex environment for the generation of oceanic kinetic energy (Smith, 2007). Given that, the characteristics of the kinetic energy cascade induced by baroclinic instability in this region—and its influence on the bifurcation point of the bidirectional cascade—remain unclear.

Despite theoretical predictions, modeling studies in this region have produced divergent results regarding the bifurcation scale of bidirectional kinetic energy transfer. Ajayi et al. (2021) identified the bifurcation point at approximately 20 km using a 1-km resolution model, while Srinivasan et al. (2023) reported a much smaller bifurcation scale of about 5 km based on a 500-m resolution model. These differences likely stem from the use of different model configurations and resolutions, highlighting the need for observational data to constrain the kinetic energy transfer characteristics in this region.

In the subpolar region of the North Atlantic, high spatial resolution Shipborne Acoustic Doppler Current Profiler (SADCP) data has been accumulated since 1999 (Kuntsen et al. 2005). These measurements offer high-resolution velocity data with a spatial resolution of 1–3 km, providing sufficient coverage to capture velocity characteristics across a range of scales, from mesoscale to submesoscale. The spatial distribution of velocity data provided by SADCP is a crucial tool for diagnosing the kinetic energy cascade (Qiu et al. 2022). Based on SADCP data, this paper analyzes the general characteristics and seasonal variations of kinetic energy cascades in the Iceland

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Basin and Irminger Sea. Energy injection scales are identified at approximately 4–8 km and 13 km in both the Iceland Basin and the Irminger Sea, with the dominant injection peaks exhibiting seasonal modulation in intensity rather than in scale. These injections are particularly intense during spring and fall, aligning with enhanced submesoscale kinetic energy and inverse cascade processes, while all these phenomena weaken in summer. The paper proceeds with descriptions of the datasets and methods in section 2, followed by the main results in section 3. The summary and discussion are presented in section 4.

2. Data and Method

a. Data

1) Shipborne Acoustic Doppler Current Profiler velocity data

The upper ocean velocity data analyzed in this study were collected by shipboard Acoustic Doppler Current Profilers (SADCPs) mounted on the vessel Nuka Arctica, which operates on a regular three-week schedule between Greenland and Scotland, occasionally stopping in Iceland. The dataset includes two components: an earlier 150 kHz ADCP operated between 1999 and 2002, providing profiles to ~400 m depth, and a later 75 kHz ADCP used from 2012 to 2019, extending coverage to ~800 m. Data processing and quality control procedures are detailed in Knutsen et al. (2005), Chafik et al. (2014), and Childers et al. (2015). These datasets have been previously used to quantify poleward volume, heat, and freshwater fluxes in the eastern subpolar North Atlantic (Rossby et al., 2017).

The vessel travels at a typical speed of 8m/s, covering ~5 km in 11 minutes, or ~100 km in 3.5 hours. With a 5-minute averaging interval, the along-track spatial resolution of the SADCP data is approximately 2.5 km. The climatological mean buoyancy frequency in the upper ocean is about 7.5×10^{-6}, corresponding to a free internal gravity wave period of roughly 5 hours. Internal waves with wavelengths under 50 km, as well as mode-1 internal tides with a phase speed around 1.6 m/s (based on eigenfunction solutions), propagate more slowly than the vessel. This supports the

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validity of the fast-tow approximation used in this study (Rocha et al., 2016b; Soares et al., 2022).

The shipboard ADCPs (SADCPs) measure water velocity relative to the ship, and to obtain geographically referenced velocities, the ship's own motion must be removed. This correction relies on GPS-based navigation and heading measurements, with vessel speed affecting along-track components and heading accuracy determining cross-track components. Further details on instrument installation, data acquisition, and processing can be found in Chafik et al. (2014). Despite frequent inclement weather in the subpolar North Atlantic, the SADCP data have been shown to be of high quality (Chafik et al., 2014; Rossby et al., 2017), owing in part to the vessel's deep (8 m) draft and the recessed bow thruster design, which minimizes air entrainment under rough conditions.

According to the uncertainty analysis in Chafik et al. (2014), the instrumental error due to heading inaccuracy (~0.05°) corresponds to a horizontal velocity uncertainty of approximately 0.004 m/s. Cross-track biases between successive transects are about 0.0022 m/s. These estimates indicate that residual systematic and instrumental velocity errors are on the order of 0.002–0.004 m/s, ensuring the reliability of the SADCP dataset for upper-ocean current analysis.

To assess the impact of instrumental uncertainty on the estimated structure functions, we assume uncorrelated Gaussian errors in the measured velocities, with a standard deviation of approximately 0.004 m/s for both longitudinal and transverse components, based on the SADCP system performance reported by Chafik et al. (2014). For the second-order structure function, the contribution of measurement noise results in an additive bias of roughly 10^{-6} m²/s². This value is negligible when compared to the typical magnitudes of observed second-order structure functions, which are on the order of 10^{-2}m²/s². For the third-order structure function defined as the product of longitudinal velocity increment and the sum of squared longitudinal and transverse increments, the propagated uncertainty due to measurement error is estimated to be 10^{-7} m³/s³. This corresponds to a relative error of less than 10 percent. These calculations confirm that instrumental errors have a minimal influence on the estimation

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of both second- and third-order structure functions in this study. For detailed estimation, see Appendix A.

To ensure data quality, we applied a multi-step quality control procedure to the velocity measurements across all transects. This included visual inspection, removal of outliers beyond three standard deviations under the assumption of Gaussian-distributed errors.

After quality control of the SADCP velocity data, the final quality-controlled dataset includes 109 transects in the Iceland Basin and 84 transects in the Irminger Sea as shown in Figure1. For all structure function calculations and spectral diagnostics in this study, the SADCP velocity data were vertically averaged from the surface to 400 m. This depth was chosen to ensure consistency across the entire dataset, as the earlier SADCP observations from 1999 to 2002 only extended to a maximum depth of approximately 400 m. Sensitivity tests using shallower averaging depths of 300 m and 200 m showed no substantial effect on the distribution pattern of positive and negative values. Therefore, the main results presented here are not sensitive to the specific choice of averaging depth.

To isolate mesoscale and submesoscale fluctuations from the large-scale background flow, we first interpolated all shipboard velocity transects onto a uniform 2.5 km longitudinal grid. A meridional mean velocity profile was then calculated by averaging across all transects within each region. This mean profile was subtracted from each transect to remove the mean flow, following the approach of Qiu et al. (2022). After this background subtraction, velocity differences were computed between all pairs of points within each transect, and their corresponding spatial separations were recorded to construct a velocity-difference database. These pairwise differences were subsequently aggregated across all transects to calculate second- and third-order structure functions. To assess statistical uncertainty at each spatial separation scale, we applied a bootstrap resampling method with 2000 realizations, and calculated the standard error accordingly. The resulting confidence intervals are shown as error bars in the following figures.

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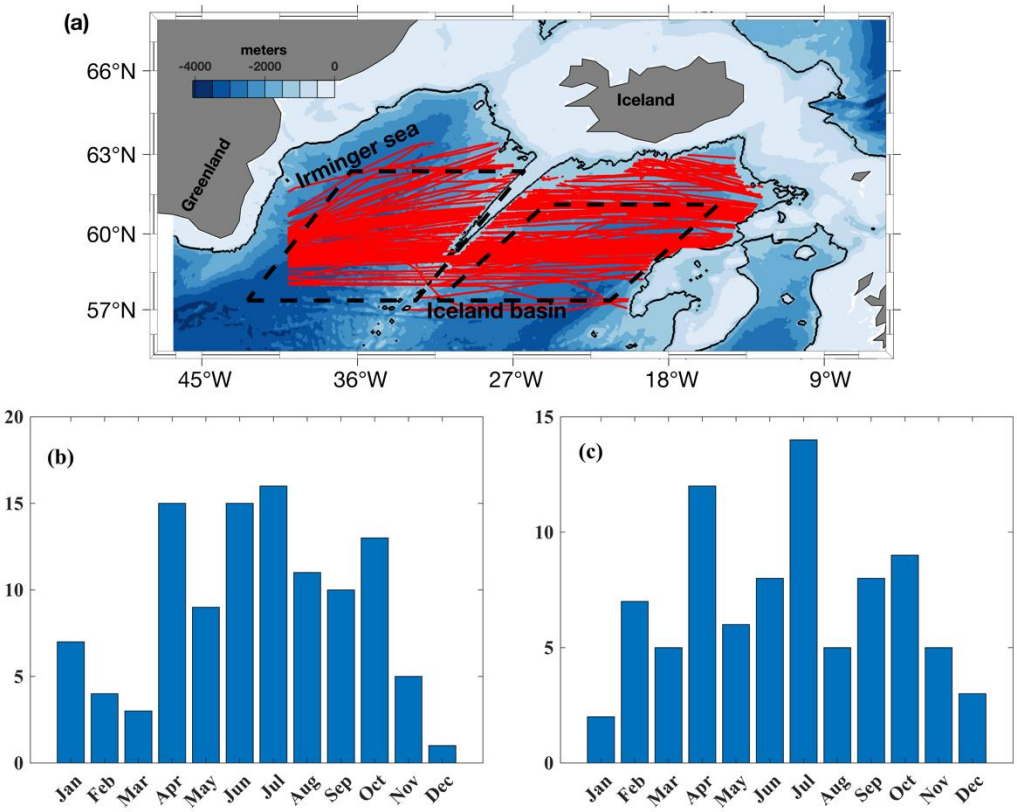

Figure. 1. (a) Routes for vessel Nuka Arctica in the subpolar North Atlantic. They correspond to the ADCP transects used in this study. The black dashed line indicates the area where EKE was calculated, as shown in Figure 4. The total number of ADCP transects is aggregated into 12 months for the Iceland Basin (b) and Irminger Sea (c). The black solid line in (a) denotes the 1200m isobath.

2) WOA data

We utilize temperature, salinity, and velocity data from the World Ocean Atlas (WOA) climatology dataset (https://www.ncei.noaa.gov/access/world-ocean-atlas-2023/bin/). WOA data is used to compute the linear instability characteristic scales, which are subsequently compared with the kinetic energy injection scales derived from the SADCP data.

To provide hydrographic and dynamical context for the study region, Figure 2 illustrates the large-scale potential density structure and seasonal evolution of mixed layer depth (MLD) across the Irminger Sea and Iceland Basin. Figure 2(a) shows a zonal section of potential density along approximately 60°N, with the black line denoting the climatological MLD and the vertical dashed line indicating the boundary between the two regions.

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These hydrographic contrasts provide important context for understanding the scale-dependent energy cascade characteristics explored in the subsequent sections. In both basins, strong horizontal density gradients are observed along the periphery, largely shaped by topographic influences. These gradients are associated with the boundary currents that circulate around the Iceland Basin and the Irminger Sea. In the long-term mean, the mixed layer depth (MLD) generally remains around 150 m. However, as further illustrated in Figure 2(b), the MLD exhibits pronounced seasonal variability. In March, the MLD reaches its maximum—up to 600 m—in the Irminger Sea, slightly deeper than in the Iceland Basin. During July, both basins exhibit a significant shoaling of the mixed layer, with minimum depths of approximately 20 m. The mixed layer depth is often considered a key indicator of the available potential energy reservoir that supports the development of submesoscale processes, which in turn influence mesoscale eddies and other dynamical features through kinetic energy cascades (e.g., Fox-Kemper et al., 2008; Rocha et al., 2016b). Therefore, in the following sections, we focus on investigating the kinetic energy cascade in this region, with particular attention to its seasonal variability.

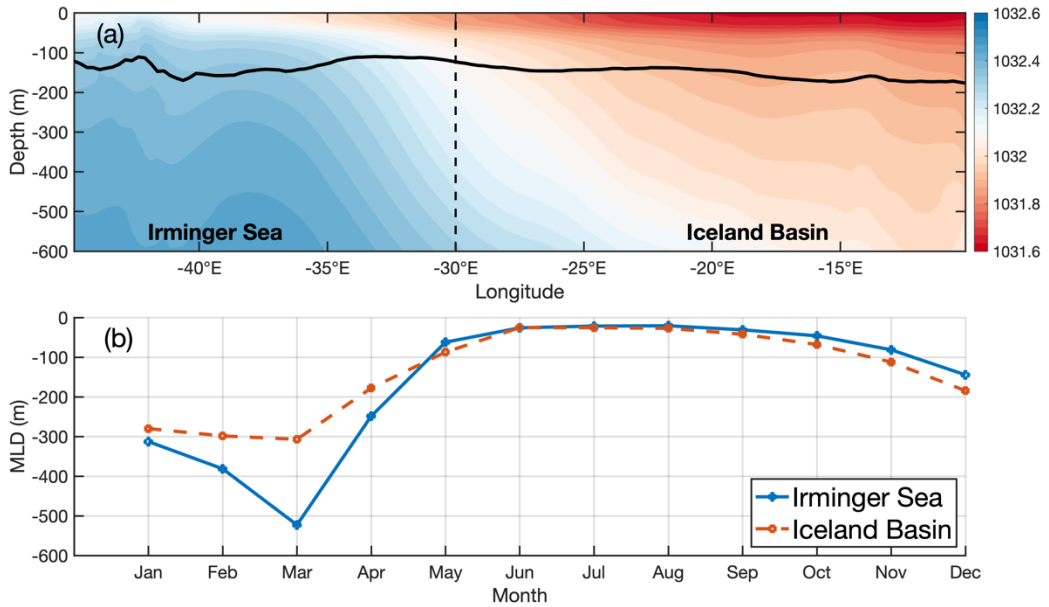

Figure 2 (a) Climatological meridional section of potential density at 60°N across the Iceland Basin and Irminger Sea. The black line indicates the corresponding mixed layer depth. (b) Multi-year seasonal averages of mixed layer depth in the two study regions

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(as defined by the red boxes in Figure 1), showing the seasonal variation of MLD in the Iceland Basin and Irminger Sea.

3) Sea surface height

Surface geostrophic velocity fields between 1998 and 2019 are downloaded from the Copernicus Marine and Environment Monitoring Service (CMEMS; https://data.marine.copernicus.eu/product/SEALEVEL_GLO_PHY_L4_MY_008_047/description). The along-track sea surface height anomalies were used to calculate geostrophic velocities following geostrophic balance. The eddy kinetic energy is defined as $EKE = \frac{1}{2}\left(u'^2 + v'^2\right)$, where u' and v' are derived by removing the long-term mean from the total surface geostrophic velocity.

b. Method

1) Velocity structure function

For each shipboard tracks, we first remove the mean value of the velocity. We then compute the difference in along-track velocity (u) and across-track velocity (v) between each pair of observation points for each shipboard track, denoted as $\delta u(r)$ and $\delta v(r)$, where r represents the straight-line distance between two points. Based on these velocity differences $\delta u(r)$ and $\delta v(r)$, the second-order structure function D2(r) (Davidson, 2015) was calculated as

$$\mathrm{D2(r)} = \delta u^2(r) + \delta v^2(r)$$

For isotropic and homogeneous turbulence, the second-order structure function $\mathrm{D2(r)}$ is closely related to the kinetic energy spectrum E(k) (Davidson, 2015). When the energy spectrum follows a power-law form $E(k) \sim k^{-m}$, where k is wavenumber, this relationship implies that $D2(r) \sim r^{m-1}$ in physical space, after transforming from wavenumber to spatial separation (Lindborg, 2007).

In isotropic and homogeneous oceanic turbulence, two characteristic slopes of the kinetic energy spectrum, -3 and -5/3, are associated with different theoretical regimes. A slope of -3 corresponds to the enstrophy cascade predicted by interior quasi-

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geostrophic (QG) turbulence theory (Charney, 1971), while a slope of -5/3 is consistent with the kinetic energy cascade described by surface quasi-geostrophic (SQG) turbulence theory (Blumen, 1978; Callies and Ferrari, 2013). In idealized QG, the inverse kinetic energy and forward enstrophy cascades are expected to occupy distinct inertial ranges separated by the energy injection scale. However, in the oceanic context—especially under non-stationary forcing and multi-scale baroclinic instabilities—this separation can become blurred. Energy may be injected across a spectrum of scales rather than a single wavenumber, resulting in partial overlap between the inverse and forward cascade ranges. This overlap manifests in intermediate spectral slopes between -5/3 and -3, consistent with previous observational findings (e.g., Chai and Zhao, 2024).

Therefore, in this region, the second-order structure function $D2(r)$ is expected to exhibit characteristic slope between 2 and 2/3.According to turbulence theory, both forward and inverse kinetic energy cascades exhibit the same slope of 2/3 in $D2(r)$, making it necessary to use $D3(r)$ for further diagnosis the direction of energy cascade (Lindborg and Cho 2001). At scales where we observe slopes close to 2/3, the signal may partly reflect the contribution of the internal wave continuum. Rocha et al. (2016a) demonstrated that in the Subpolar North Atlantic, kinetic energy spectra often show evidence of internal wave contamination, particularly in the slope range close to 2. For the corresponding second-order structure function, this translates into a slope close to 1. This suggests that a portion of the variance detected in our SADCP data could be influenced by internal wave motions, in addition to turbulent dynamics.

The third-order structure function is defined as:

$$D3(r) = \delta u(\delta u^2(r) + \delta v^2(r))$$

For isotropic and homogeneous flows, according to Kolmogorov's theory, in the inertial subrange (where energy transfers away from injection scales), the third-order structure function D3(r) follows the scaling relation for third-dimensional turbulence:

$$D3(r) = \overline{\delta u^3} = -\frac{4}{5}\ \epsilon r$$

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where ϵ is the energy dissipation rate. On the other hand, for two-dimensional turbulence, Kraichnan (1967) further refined the theory, suggesting that the energy dissipation satisfies the following relationship:

$$\mathrm{D3(r)} = \delta u^3 = \frac{3}{2}\epsilon \mathrm{r}$$

Therefore, the slope of $\mathrm{D3(r)}$ is first-order function of the distance r for kinetic energy cascades, regardless of two-dimensional and third-dimensional turbulence. In other words, when the slope of the third-order structure function approaches 1, this scaling behavior serves as a diagnostic for determining the direction and nature of the kinetic energy cascade. When $\mathrm{D3(r)}$ >0 indicates the inverse kinetic energy cascades, the kinetic energy transfers from small-scale ocean processes to large-scale ones. When $\mathrm{D3(r)}$ <0 represents a forward cascade of kinetic energy, the kinetic energy is transferred from large scale to small scale.

However, there are additional complexities to consider to distinct the direction of energy cascade. In two-dimensional turbulence, enstrophy is also a conserved quantity, and it can transfer across scales from large scale to small scale. This enstrophy cascade is reflected in the third-order structure function $\mathrm{D3(r)}$ as well. Forward enstrophy cascade obey:

$$\mathrm{D3(r)} = \delta u^3 = \frac{1}{8}\eta \mathrm{r}^3$$

where η indicate the dissipation rate of enstrophy. At submesoscale to mesoscale ranges, oceanic motions are constrained by Earth's rotation and stratification, which tend to make the flow behave more like quasi-two-dimensional turbulence. As a result, the slope of the third-order structure function in this range typically falls between 1 and 3. This makes it difficult to determine the kinetic energy cascade directions based on the third-order structure function directly. In addition, the slope determination of the third-order structure function requires the study scale limiting in the inertia scale, that is, the scale away from the original kinetic energy injection scale. However, there are many instability processes in the submesoscale to mesoscale interval, making it difficult to determine the kinetic energy cascade direction directly by using the third-order

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structure function. These factors also lead to the slope of the third-order structure function sometimes deviating from 1-3 (e.g., Qiu et al. 2022).

2) Energy flux and injection

In order to find out the actual direction of the kinetic energy cascade and the characteristics of energy injections, we use the diagnostic methods based on D3(r) proposed by Xie and Buhler (2019). By applying a Fourier transform, the third-order structure function is converted to the wavenumber spectrum, expressed as:

$$\mathrm{V(r)} = -4r\int_0^{\infty}\frac{1}{k}Fr(K)J_2(K)dk$$

where J_2 is the second-order Bessel function, $Fr(K)$ is the energy forcing function, and K is the wavenumber. For energy injection at a single scale, $Fr(K)$is written as:

$$Fr(K) = -\epsilon_u + \ \epsilon H\left(K - K_f\right)$$

Here, ϵ_u represents the energy cascade from the injection scale to larger scales, K_f is the energy injection scale, and ϵ is the energy injection rate at the injection scale. The forward energy cascade is represented by$\epsilon_d = \epsilon - \epsilon_u$, and energy dissipation is assumed to occur at infinitely large and infinitely small scales. Next, $Fr(K)$ is expressed as a linear superposition of energy injection scales:

$$Fr(K) = -\epsilon_u + \ \sum_{j=1}^{N_f}\epsilon_j H\left(K - K_j\right)dk_j$$

Substituting this into the equation, the third-order structure function becomes:

$$\mathrm{V(r)} = 2\epsilon_u r - \ \sum_{j=1}^{N_f}4\frac{\epsilon_j}{k_j}J_1\left(rK_j\right)dk_j$$

where$J_1$is the first-order Bessel function. Using shipboard ADCP data to measure velocity, we obtained the third-order structure function $\mathrm{V(r)}$. We then solved for ϵ_j and k_j through fitting, which allowed us to study energy transfer and injection (Balwada et al. 2022).

Our approach shares methodological similarities with previous studies in the estimation of energy injection scales from third-order structure functions (Balwada et al. 2022). Specifically, we employed both the least-squares method (LS) and the non-

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negative least-squares (NNLS) method to fit the observed structure function V(r). Among these, the results obtained using NNLS are presented in the main text, as they offer a physically consistent estimate of the dominant energy injection scales by enforcing the constraint that energy flux contributions are non-negative. This regularization reduces the risk of overfitting to noise and allows for clearer physical interpretation of the flux distribution. In contrast, the unconstrained least-squares results, which are more sensitive to data variability and may produce physically ambiguous estimates, are included in the Appendix C to facilitate comparison with analyses that do not incorporate regularization or with coarse-graining approaches.

3)Local linear instabilities analyses

To verify the reliability of the energy injection scales derived from the third-order structure function (i.e. the ones have largest linear growth rates), we calculate local baroclinic growth rates and identify the scales of the fastest-growing modes, following the method used in Smith (2007). Under the assumption of a rigid lid and a flat bottom, the quasi-geostrophic potential vorticity equation is obtained,

$$\partial_t q + J(\psi, \beta y + \Gamma\psi) = 0, -H < z < 0$$

$$\partial_t b + J(\psi, b) + J(\psi, f\partial_z\psi) = 0, z = -H, 0$$

where $\Gamma = \partial_z(f^2/N^2)\partial_z$, $b = f\partial_z\psi$, $q = (\nabla^2 + \Gamma)\psi$ is the qusi-geostrophic potential vorticity. Assuming that the flow function satisfies the wave solution, $\psi = \phi(z)e^{i(kx+ly-\omega t)}$, k and l are horizonal wavenumber, ω is the eigenvalue. When currents velocity and stratification characteristics are known we can solve it in the form of the characteristic function to obtain the most unstable scale, by $r = (k^2 + l^2)^{-\frac{1}{2}}$. As discussed by Smith (2007) and Tulloch et al. (2011), this method can identify not only first-order baroclinic instability scales, but also instability above the thermocline, including mixed-layer instability and Charney-like baroclinic instability. These instabilities have the ability to convert available potential energy (APE) into kinetic energy (KE) in the upper ocean. Therefore, we believe that this method can provide a

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good comparison with the energy injection scale obtained by the Xie and Buhler (2019) method based on the third-order structure function.

4) Spectral estimation and band-pass integration

We compute along-track 1-D kinetic-energy spectra $\phi_{KE}(k)$ from detrended SADCP velocities using Welch averaging with a Hanning window. To quantify the variance associated with specific scale ranges, we apply a wavenumber-space band-pass by integrating the spectrum over prescribed bands $[k_1, k_2]$:

$$E_{[k1,k2]} = \int_{k1}^{k2} \phi_{KE}(k)dk$$

This operation is equivalent to a spatial band-pass filter and is used to separate mesoscale and submesoscale contributions to kinetic-energy variance.

3.Result

a. Second-order structure functions and third-order structure functions

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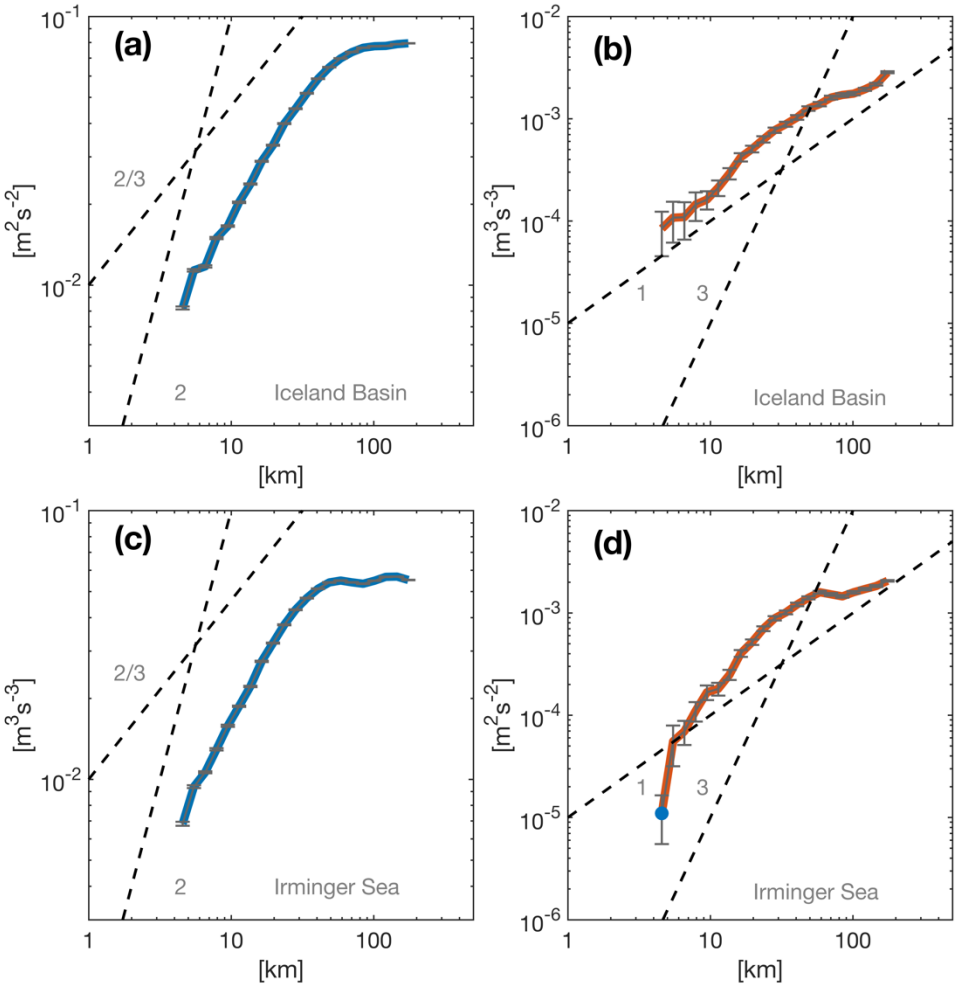

Figure. 3 Mean two-order structure function(D2(r); blue lines) and third-order structure function(D3(r); red lines) in Iceland Basin(a)(b) and Irminger Sea(c)(d). Error bars represent the standard error estimated from 2000 bootstrap samples.

In the Iceland Basin, all available SADCP data are used to calculate the second- and third-order structure functions, D2(r) and D3(r). Within the 4 – 100 km scale range, the slope of D2(r) is approximately 0.71. This value lies between the characteristic slopes of 2/3 (associated with the forward kinetic energy cascade) and 2 (associated with the enstrophy cascade), indicating that both cascades coexist in this region. Beyond 100 km, the energy spectrum flattens and the slope drops below the theoretical minimum of 2/3, suggesting that velocity fluctuations become decorrelated at larger scales—likely exceeding the dominant eddy sizes. This interpretation is supported by Appendix C, which shows the loss of velocity correlation at these scales. Therefore, the analysis in the Iceland Basin is constrained to scales below 100 km,

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where the assumption of isotropy is more valid and structure functions exhibit turbulence-like scaling.

To further investigate the energy cascade direction, the third-order structure function D3(r) is analyzed. Within the 4–100 km scale range, its slope is approximately 1.11, again lying between the slopes of 1 (kinetic energy cascade) and 3 (enstrophy cascade). This may indicate the simultaneous presence of both cascades, however, the lack of alignment with canonical slopes suggests that existing theoretical frameworks may be insufficient to fully capture the dynamics observed. D3(r) remains positive across all scales in the Iceland Basin, indicating no evidence for a forward (downscale) kinetic energy cascade.

In the Irminger Sea, the slope of D2(r) is 0.84 over the 4–70 km scale range, also reflecting a mixture of both kinetic energy and enstrophy cascades. Similar to the Iceland Basin, the energy spectrum flattens beyond 70 km, with the slope dropping below 2/3. Hence, the analysis in this region is limited to the 4–70 km range, where isotropy and turbulence-like behavior are more applicable.

For the third-order structure function D3(r), the slope is approximately 1.68 across the 4–70 km range, again falling between the theoretical slopes for kinetic energy (1 slope) and enstrophy (3 slope) cascades. Unlike in the Iceland Basin, the Irminger Sea exhibits a positive forward cascade signal at around 4 km, which may suggest the presence of a kinetic energy cascade bifurcation point. However, this signal lies near the edge of the third-order structure function, and its physical relevance is uncertain. As will be discussed in Section2b, further inspection reveals that this feature is likely driven by wintertime sampling biases.

The upper bounds used to define the scale range of isotropic turbulence are set to 100 km for the Iceland Basin and 70 km for the Irminger Sea. Although the scale range for turbulence analysis differs between the two basins, sensitivity tests confirm that selecting an upper bound between 50 km and 150 km does not significantly alter the results of this study in either region.

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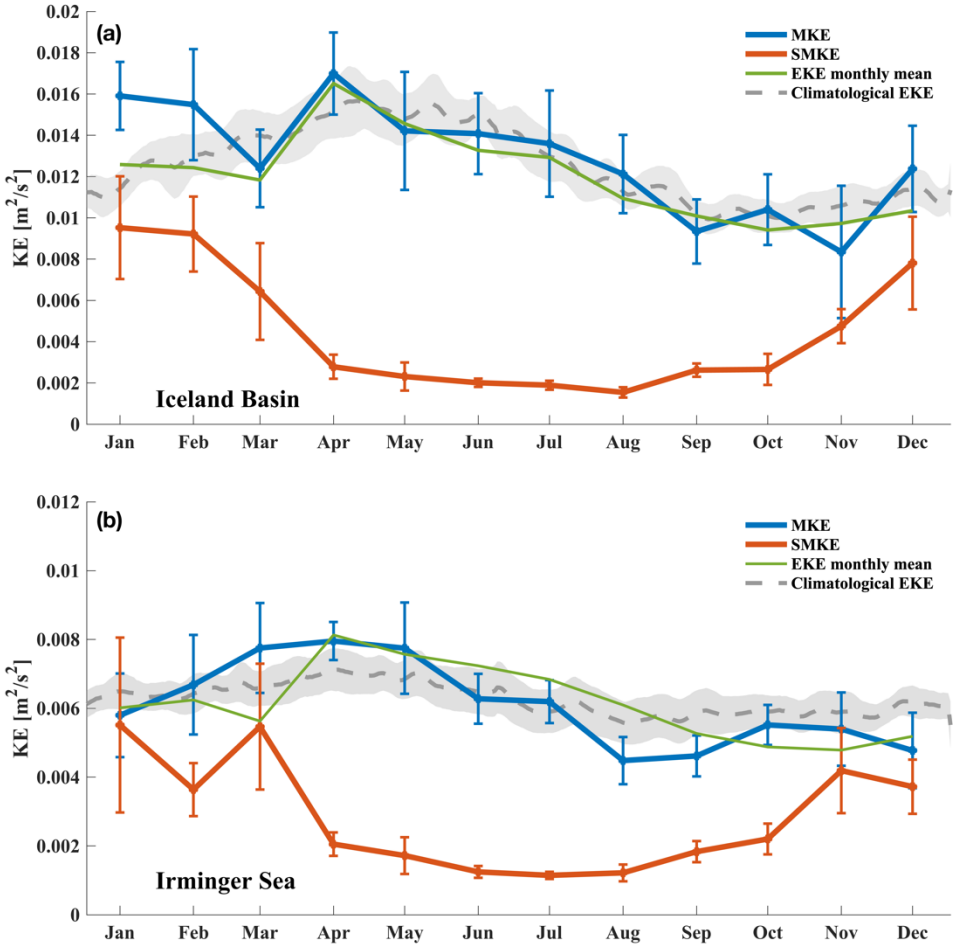

Figure 4. Kinetic energy (KE) from SADCP data aggregated over different scale ranges for (a) the Iceland Basin and (b) the Irminger Sea. The blue line represents mesoscale kinetic energy (MKE), obtained by band-pass integrating the SADCP-derived KE spectra over 40–150 km in the Iceland Basin and 40–100 km in the Irminger Sea. The orange line shows submesoscale kinetic energy (SMKE), estimated from spectral integration at scales smaller than 40 km in both basins. The green line shows the climatological mean eddy kinetic energy (EKE) derived from AVISO data, matched in location and time to the SADCP observations. The gray line indicates the seasonal variation of climatological EKE in each region, also based on AVISO data. The spatial range used for EKE estimation is defined by the black dashed boxes shown in Figure 1.

Based on the above analysis, we compute the band-limited mesoscale kinetic energy (MKE) by integrating the SADCP-derived kinetic energy spectra over the wavenumber ranges corresponding to 40–150 km in the Iceland Basin and 40–100 km in the Irminger Sea (Figure 4, blue lines). It is important to note that our primary focus

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is on the seasonal modulation of MKE and submesoscale kinetic energy (SMKE), rather than on a strict dynamical classification of these scales. The choice of 40 km as the lower bound for the mesoscale range is motivated by the fact that eddy signals with radii of approximately 40 km can be resolved from along-track satellite altimetry data (Chelton et al., 2011), allowing for consistent comparisons with ship-based SADCP observations. While this threshold does not represent a strict dynamical boundary, it captures the smallest mesoscale features reliably detected by both platforms. At this scale, the mesoscale kinetic energy (MKE) exhibits pronounced seasonal variability, peaking in March and gradually declining through September in the Iceland Basin and August in the Irminger Sea, after which the energy begins to intensify again.

To explain the seasonal variations in MKE, we use the seasonal variation of eddy kinetic energy (EKE) calculated from satellite altimeter data. Both basins exhibit significant seasonal variations, with EKE peaking in spring (March–May) and reaching its lowest values in fall (September–November).

Apart from a few instances—such as in the Iceland Basin during January and February—mesoscale kinetic energy (MKE) derived from SADCP observations generally agrees well with the eddy kinetic energy (EKE) estimated from satellite altimetry, confirming the reliability of our observational data. MKE exhibits a clear seasonal cycle, peaking in April, followed by a steady decline and a slight rebound during November–December.

This monthly variability pattern differs significantly from that of submesoscale kinetic energy (SMKE, defined here as the kinetic energy integrated over horizontal scales from 4 to 40 km), which reaches its maximum in both basins during January–March, drops to a minimum during June–August, and then increases again after October. Although we acknowledge that observational uncertainties in the Iceland Basin during January–February may affect both MKE and SMKE estimates, the strong SMKE observed in March is considered robust. This is supported by the fact that March corresponds to the period of deepest mixed layers throughout the year—a condition commonly associated with enhanced submesoscale activity.

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The distinct seasonal patterns of MKE and SMKE raise questions about the underlying energy transfer mechanisms across scales. In particular, it is of interest to assess whether these patterns are consistent with the inverse kinetic energy cascade framework established in lower-latitude regions (Dong et al., 2020; Schubert et al. 2023), which predicts a one- to two-month lag between submesoscale KE peaks and subsequent increases in mesoscale EKE (here referring to the EKE values derived from satellite data).

In our study region, this relationship appears to hold during spring: as shown in Figure 4a–b, SMKE peaks in March, followed by elevated EKE in April and May—consistent with a delayed inverse cascade from submesoscales to mesoscales. However, this pattern does not persist into fall. From September to November, SMKE increases substantially, yet mesoscale EKE remains weak and does not exhibit a corresponding delayed enhancement. This seasonal contrast may be linked to differences in the direction of kinetic energy cascades across scales. To explore this possibility, we further diagnose the seasonal variability of energy cascade pathways in the following section.

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b. The seasonal variations of second-order function third-order function

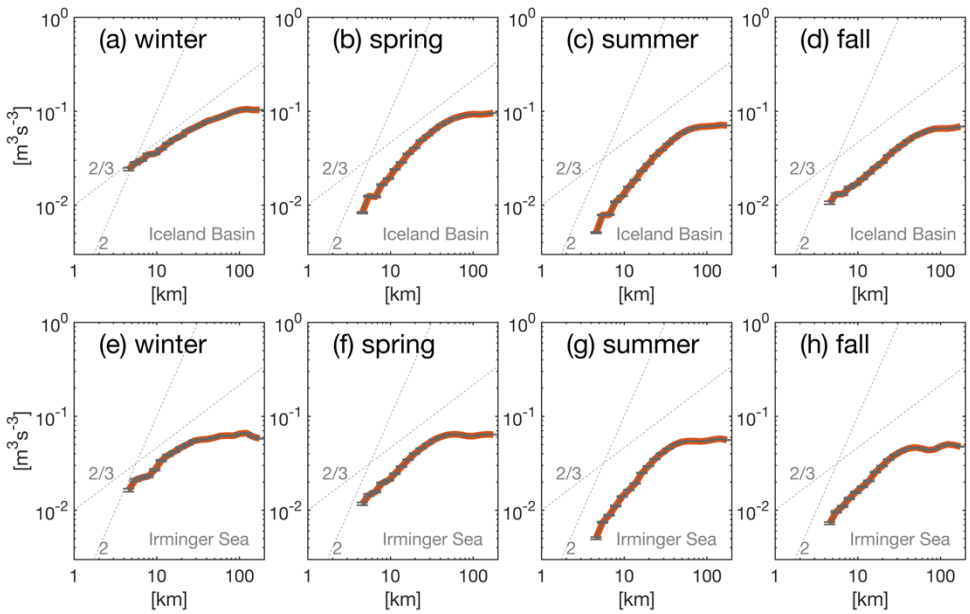

Figure 5 Second-order structure functions of horizontal velocity differences as a function of separation distance for different seasons. (a–d) show results from the Iceland Basin during winter, spring, summer, and fall, respectively. (e–h) show corresponding results for the Irminger Sea. Second-order structure functions are plotted with logarithmic scales in both axes. Error bars represent the standard error estimated from 2000 bootstrap samples. Reference slopes of 2/3 and 3 are shown for comparison.

We first performed a seasonal decomposition of the second-order structure function. In this study, we define winter as December–February, spring as March–May, summer as June–August, and fall as September–November. Within the study domain, the slope in winter is consistently smaller than 2/3, which is the characteristic slope predicted by the theoretical solution. After applying quality control, the number of valid wintertime samples was limited to 11 in the Iceland Basin and 9 in the Irminger Sea.

For winter, although the computed slopes of D_2 are not far from theoretical predictions (2/3 slope), the number of available SADCP sections is too limited and the measurements are less stable compared with other seasons. As a result, the winter estimate is not robust enough to provide a reliable diagnostic value. We therefore focus our seasonal interpretation on spring, summer, and fall. Nonetheless, smaller eddies are more common in winter, which may partly explain the deviations observed.

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The second-order structure function exhibits clear seasonal variability, consistent with the kinetic energy spectra previously reported for this region (Chai and Zhao 2024). In the Iceland Basin during spring, and in both basins during summer, the slope approaches 2, indicating characteristics of enstrophy cascade. In contrast, during other seasons, the slope is closer to 2/3, which is more consistent with a classical kinetic energy cascade. However, the direction of the kinetic energy cascade cannot be determined based on the second-order structure function alone. Therefore, we further diagnose the third-order structure function to infer the directionality of energy transfer.

As shown in figure 6, in the Iceland Basin, the slope of the third-order structure function D3(r) over the 4–100 km scale range is approximately 1.1 in winter, significantly higher than values observed during other seasons across all scales. In spring, the slope increases to 2.2, suggesting the coexistence of both enstrophy and kinetic energy cascades. During summer, the slope further increases to 2.7, approaching the theoretical value of 3, which is indicative of a dominant forward enstrophy cascade. In contrast, the slope decreases to 1.6 in fall, aligning more closely with the theoretical slope of 1, and implying a stronger contribution from the inverse kinetic energy cascade. Notably, D3(r) remains positive throughout all seasons, reinforcing the persistent presence of lack of a forward cascade in this region.

In the Irminger Sea, D3(r) exhibits distinct seasonal variations that reflect shifts in energy cascade regimes. In winter, values at scales below 10 km become negative, suggesting a forward kinetic energy cascade at submesoscales. However, the slope flattens at these scales, likely due to noise-induced errors in the third-order statistics. In spring and summer, the slopes over the 4–70 km range are both around 2.6, close to the theoretical value of 3, indicating a dominant enstrophy cascade and the coexistence of inverse kinetic energy transfer. In fall, the slope decreases to 1.2, pointing to a more prominent inverse kinetic energy cascade. In all non-winter seasons, D3(r) remains positive, suggesting no forward cascade is detected within the resolved scales.

When combined with the seasonal variations in EKE and submesoscale kinetic energy shown in Figure 3, we observe that higher mesoscale and submesoscale kinetic energy in spring. In summer, EKE is second only to that in spring, while the

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submesoscale kinetic energy is at the lowest level of the year. As a result, the characteristics of kinetic energy cascade are overshadowed by those of the enstrophy cascade during summer, making D3(r) steeper. In fall, weaker EKE corresponds to a diminished enstrophy cascade signal, while the increase in submesoscale kinetic energy leads to the dominance of kinetic energy cascade characteristics in D3 (r).

It is worth noting that no significant evidence of forward kinetic energy cascade—indicated by a negative D3 (r) with a slope of 1—is found in either the Iceland Basin or the Irminger Sea, as no such negative signal is observed in the D3(r) (except for potentially erroneous wintertime results in Irminger Sea). This is significantly different from the results of third-order structure functions in other regions at mid- and low-latitudes areas (Qiu et al. 2022). In the Iceland Basin and Irminger Sea, 4km does not appear to be sufficient to discern the scale at which the kinetic energy begins to dissipate in a forward cascade. However, this phenomenon needs further proof because the D3(r) is mixed with the kinetic energy cascades of slope 1 and the enstrophy cascades of slope 3. Especially in spring and summer, the characteristics of the vortical enstrophy cascade appear to have a more significant impact on the slope of D3(r). Therefore, we need to further analyze D3(r) to get more precise kinetic energy cascades and energy injections.

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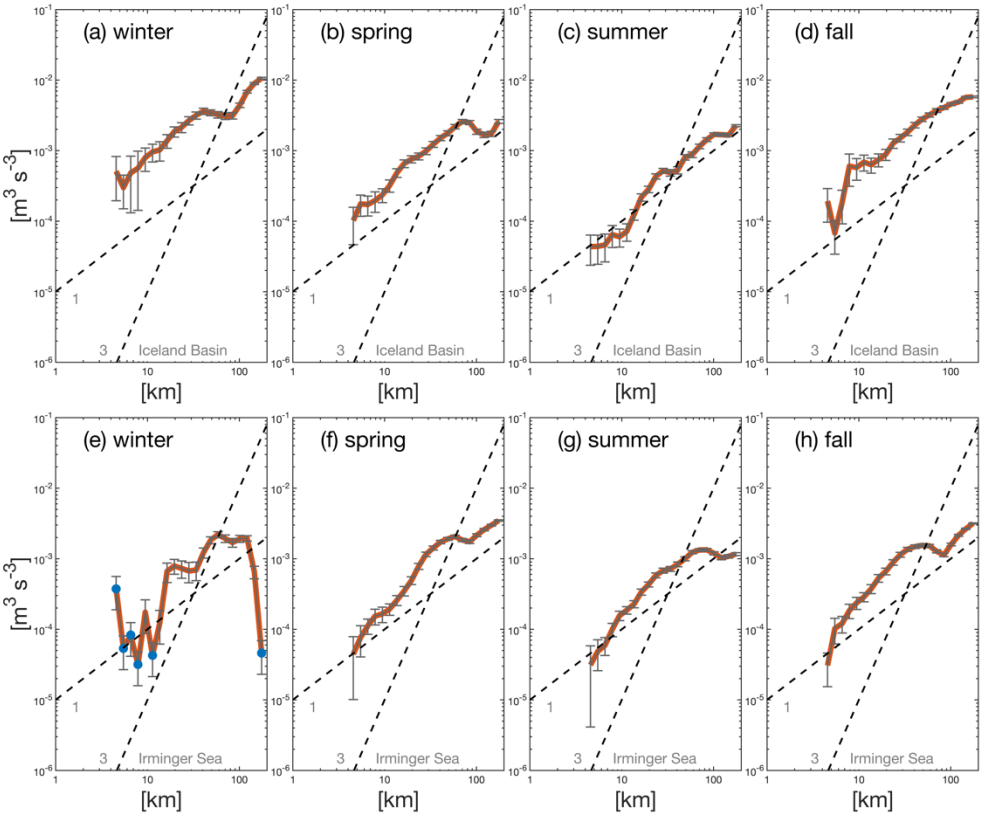

Figure 6. Mean third-order structure function using ADCP. (a–d) show results from the Iceland Basin during winter, spring, summer, and fall, respectively. (e–h) show corresponding results for the Irminger Sea. Error bars represent the standard error estimated from 2000 bootstrap samples.

c. Energy flux and injection

In order to diagnose the kinetic energy cascade and kinetic energy injection characteristics more clearly, we conduct a further analysis of the kinetic energy injection and energy cascade of the D3(r) in two regions based on the method proposed by Xie and Buhler (2019).

To illustrate the quality of the third-order structure function fitting, we include in Appendix C a set of representative examples across seasons and basins. These results confirm that the fitting procedure captures the expected scaling behavior and supports the reliability of the derived energy flux estimates.

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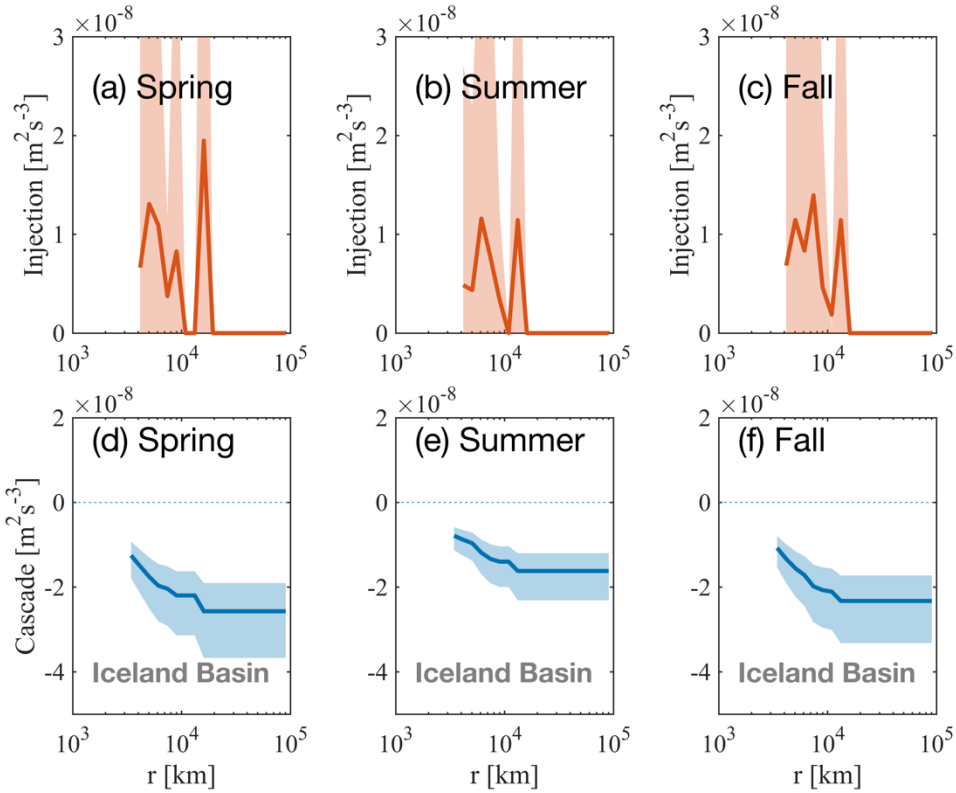

Figure 7. Mean energy injection and energy cascade in Iceland Basin using ADCP collected in spring (a, d), summer (b, e) and fall (c, f).

As shown in Figure 7, the Iceland Basin exhibits two distinct energy injection scales in each season. In spring, prominent injections occur at approximately 4–8 km and 13 km, with both contributing comparable power of approximately $1\times 10^{-8}m^2s^{-3}$ and $2\times 10^{-8}m^2s^{-3}$ respectively—representing the strongest energy injection observed throughout the year. Additionally, Figure 7 (d) shows that the Iceland Basin experiences its strongest inverse kinetic energy cascade of the year during spring, covering the entire study range from 4 km to 100 km. In summer Figure 7(b), the energy injection decreases significantly, reaching its lowest values of the year, with peaks at 6 km and 13 km. This weakest energy injection corresponds to the weakest inverse kinetic energy cascade of the year. In fall, as shown in Figure 7(c), the energy injection intensifies again, with peaks at 4-8 km and 13 km, contributing approximately $1\times 10^{-8}m^2s^{-3}$, respectively. As shown in Figure 7(f), the inverse kinetic energy cascade strengthens in fall compared to summer but remains weaker than in spring.

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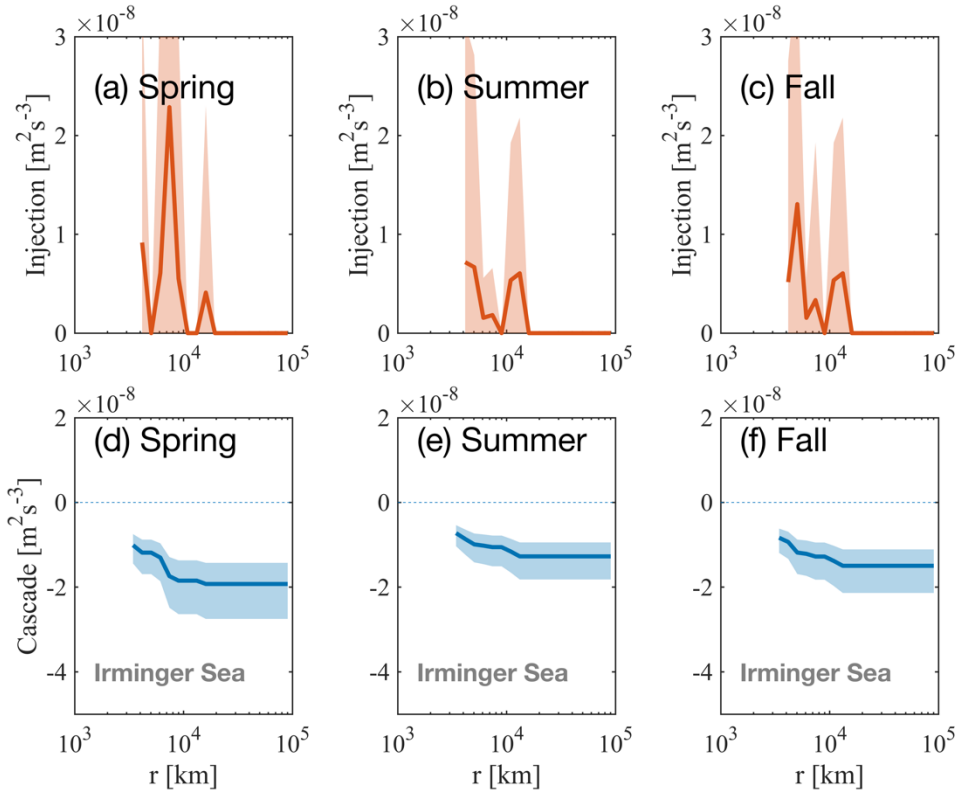

Figure 8. Mean energy injection and energy cascade in Irminger Sea using ADCP collected in spring (a, d), summer (b, e) and fall (c, f).

In the Irminger Sea, two prominent energy injection scales are identified within the study range, centered at 6–8 km and 15 km. The injection at 6–8 km corresponds to a kinetic energy input of approximately $2.3 \times 10^{-8} m^2 s^{-3}$, while the 13 km scale exhibits a much weaker input of only $0.3 \times 10^{-8} m^2 s^{-3}$. Spring features the strongest inverse kinetic energy cascade observed throughout the year. In summer, energy injection appears at both 4 km and 15 km, each contributing approximately $0.3 \times 10^{-8} m^2 s^{-3}$, coinciding with the weakest inverse cascade of the year.

In fall, energy injection is primarily concentrated at 6–8 km, with a magnitude of approximately$1 \times 10^{-8} m^2 s^{-3}$ and a weak peak at 15 km. The strength of the inverse kinetic energy cascade during this season increases relative to summer, indicating a partial recovery of upscale energy transfer.

When combined with the seasonal variations of submesoscale kinetic energy in Figure 3, it becomes evident that the intensity of energy injections and the strength of the inverse kinetic energy cascade are directly linked to submesoscale processes.

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Following the occurrence of oceanic instability (kinetic energy injections), submesoscale motions are enhanced, leading to an inverse cascade. Kinetic energy injections at the 6–8 km and 13 km scales in Iceland basin and Irminger Sea mainly, result in inverse kinetic energy cascades, with no significant evidence of forward kinetic energy cascades observed within the study range.

d. Linear analysis of instability

To further assess the robustness of our inferred kinetic energy injection scales and cascade patterns, we perform a linear instability analysis following the quasi-geostrophic framework of Smith (2007). Specifically, we solve for the most unstable vertical modes and associated wavenumbers using the World Ocean Atlas (WOA) seasonal mean hydrographic and velocity fields. As shown in Figure 9, the color shading denotes the baroclinic growth rate, while the black symbols highlight the most unstable wavenumbers (i.e., instability scales) for each season in the Iceland Basin (Figure9 a–c) and the Irminger Sea (Figure 9d–f).

In the Iceland Basin (Figure 9a–c), we identify three dominant the most unstable scales each season. During spring, the major peaks occur near ~2 km, ~6 km, and ~12 km; in summer, they shift slightly to ~2 km, ~6 km, and ~14 km; and in fall, to ~2 km, ~6 km, and ~17 km. The smallest scale, typically ~2 km, is below the resolution of our SADCP dataset and thus not directly observable in the kinetic energy injection results. However, the larger modes (near ~6 km and ~12–17 km) align well with the 4-8 km and ~13 km scales derived from the third-order structure function analysis (section 3c), reinforcing the reliability of both approaches.

A comparable pattern emerges in the Irminger Sea (Figures 9d–f). In spring, the most prominent instability scales appear near ~5 km and ~11 km, in summer near ~4 km and ~12 km, and in fall near ~3 km and ~8 km. This correspondence suggests that the kinetic energy injections are primarily driven by baroclinic instability in both basins, with multiple distinct scales contributing to the overall cascade process. In the Irminger Sea, the energy injection scales diagnosed from the third-order structure function closely match the most unstable scales predicted by linear instability analysis. However,

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this correspondence breaks down in fall, when no clear energy injection signal is detected at the 10 km scale.

Collectively, these findings indicate that baroclinic instability in the subpolar North Atlantic span multiple modes between roughly 2 km, 4–8 km and 17 km: while scales such as 4–8 km and 17 km are clearly captured by our SADCP-based analyses, others (like the ~2 km mode) remain below the current observational resolution. Moreover, the close correspondence between the linear instability framework and the D3(r) derived injection scales reinforces that baroclinic processes serve as the principal kinetic energy source in these regions, cascading kinetic energy from submesoscale to mesoscale.

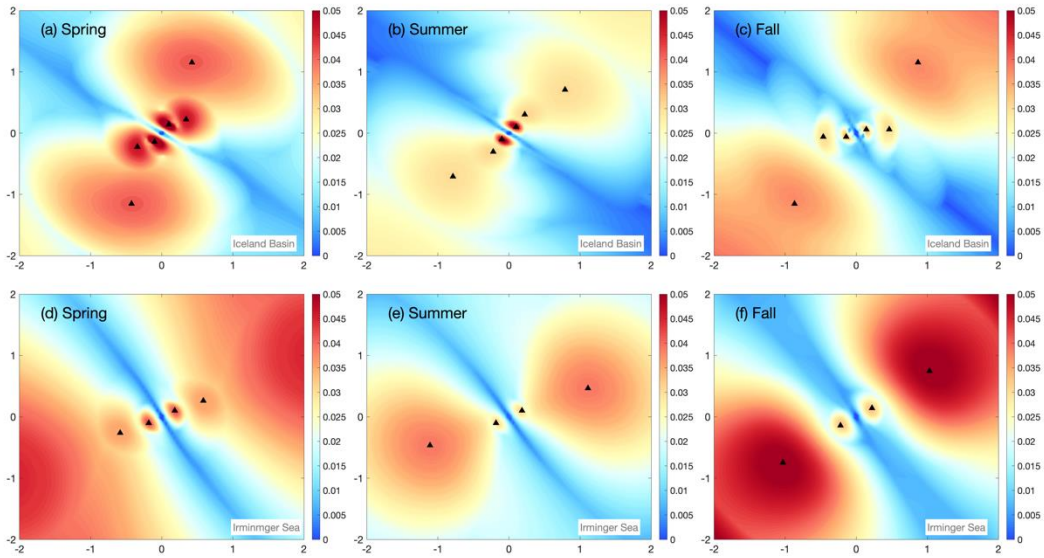

Figure 9. Local baroclinic instabilities growth rate using WOA seasonal climate mean data in Iceland Basin (a-c) and Irminger Sea (d-f).

4. Summary and Discussion

This study investigates the characteristics of energy cascades and injections in the Iceland Basin and Irminger Sea using SADCP data. We document D3(r) and, building on that, apply the method of Xie and Bühler (2019), finding that the inverse cascade dominates from the submesoscale to the mesoscale. In addition, the energy injections occur at scale of 4-8km and 13 km in Iceland Basin and Irminger. The observed seasonal variations in both kinetic energy cascades and energy injection—strongest in

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spring, slightly weaker in fall, and weakest in summer—may suggest a role for mixed-layer instabilities in the energy cascade. However, due to limitations in the available observational data, we are unable to directly diagnose their contribution.

The study of the kinetic energy cascade is based on the D3(r) analysis, which requires the assumption of isotropy. The second-order structure function, D2(r), serves as an additional constraint, with the predicted slope range for isotropic fluids lying between 2/3 and 2 according to quasi-geostrophic turbulence theory. The influence of internal waves should also be considered when interpreting our results. SADCP measurements inevitably include variability from the internal wave field. Previous studies have shown that internal waves can flatten the kinetic energy spectrum, producing slopes closer to −2 in spectral space (or equivalently D_2 slopes near 1 in physical space; Rocha et al., 2016a). Such contributions may generate intermediate slopes.

Based on this, we define 100 km and 70 km as the upper limits of isotropy for the Iceland Basin and Irminger Sea, respectively. The breakdown of isotropy at larger scales may result from various factors such as topographic influences or beta effects. At high latitudes, the larger Coriolis parameter leads to smaller Rossby deformation radii and associated dynamical scales, resulting in reduced characteristic eddy sizes compared to those typically observed in the mid- and low-latitude Pacific and Atlantic Oceans. Given the resolution of the shipboard ADCP (SADCP), this study focuses on scales ranging from 4 to 100 km in the Iceland Basin and 4 to 70 km in the Irminger Sea.

We calculate the MKE and SMKE from SADCP observations and find that both exhibit pronounced seasonal variability. MKE peaks in spring (March–April), gradually declines during summer, and shows a moderate increase again in late fall. This seasonal pattern is consistent with the variation of EKE derived from satellite altimetry, which also reaches maximum values in spring and minimum values in fall. Overall, the consistency between SADCP-derived MKE and satellite-derived EKE supports the reliability of the in-situ measurements.

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In contrast, SMKE shows a different seasonal cycle: it is strongest from January to March, reaches a minimum during June–August, and increases again in October–November. Despite some observational uncertainty in early winter, the strong SMKE observed in March aligns with periods of deep mixed layers. While the observed increase in SMKE during fall seems to coincide with a deepening of the mixed layer, the current observational dataset does not allow us to robustly establish a physical causal link between the two.

Given the distinct seasonal variations in MKE and SMKE intensity, we further analyze the seasonal behavior of D3(r). The slope of D3(r) in both basins exhibits similar characteristics, reflecting the seasonal variations of the dominant physical processes. In spring, the slope of the third-order structure function in both basins is close to 2, which lies between the canonical values of 1 (kinetic energy cascade) and 3 (enstrophy cascade). This intermediate behavior may reflect contributions from both the kinetic energy and enstrophy cascades, but could also arise from other processes such as baroclinic instabilities or internal-wave variability. During this period, both mesoscale eddy kinetic energy and submesoscale kinetic energy are strong. Mesoscale eddies correspond to the enstrophy cascade from mesoscale to submesoscale, and strong submesoscale kinetic energy corresponds to the enhancement of inverse cascades from submesoscale to mesoscale. In summer, with the rapid decline of the submesoscale processes, the enstrophy cascade caused by mesoscale eddies dominates, and the slope of the third-order structure function is closer to 3. In fall, the mesoscale eddies decrease to the lowest level, the submesoscale kinetic energy increases, and the kinetic energy cascade processes dominate the characteristics of the third-order structure function, with slope close to 1. In addition, no significant evidence of a forward kinetic energy cascade is observed at scales larger than 4 km. The increase in submesoscale kinetic energy only corresponds to the enhancement of the inverse cascade. We acknowledge that the existence of a well-defined inertial range in the ocean remains uncertain, and that slope analysis of D3(r) and D2(r) alone may be insufficient to resolve this issue.

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To further explore the characteristics of kinetic energy injections and cascades, and to rule out the possibility of the enstrophy-like cascade masking the forward kinetic energy cascade, we further diagnose kinetic energy injections and kinetic energy cascades based on Xie and Buhler (2019) method. Our results show that in Iceland Basin and Irminger Sea, there are two significant kinetic energy injection scales: one near 13 km and another at 4-8 km. The seasonal variations of kinetic energy injection scales are small, and its seasonal variation is mainly reflected in the intensity of kinetic energy injection. The intensity of kinetic energy injections is strongest in spring, slightly weaker in fall, and weakest in summer. This trend is consistent with the behavior of the inverse kinetic energy cascade.

One significant observation is the lack of a bifurcation point for the bidirectional kinetic energy cascade above the 4 km scale. Linear instability analysis reveals a notable contrast between the two basins: in the Iceland Basin, there are three significant kinetic energy injection scales, namely 2 km, 6 km, and 12-17 km, whereas in the Irminger Sea, the three significant injection scales are 3km, 4-8 km and 11-12 km. Due to the spatial resolution limitations of the SADCP data, the kinetic energy injection scales of 2km in the Iceland Basin and 3 km in the Irminger Sea cannot be resolved. However, the most unstable scales based on linear instability analysis—4-8 km and 12-17 km in the Iceland Basin and 4-8 km and 11-12 km in the Irminger Sea—are consistent with the results obtained from the SADCP data. On one hand, this confirms the reliability of the results; on the other hand, it suggests that the current resolution is insufficient to fully capture the complete kinetic energy cascade structure in the subpolar region of the North Atlantic. Higher spatial resolution observations and more accurate model simulations will be required in future studies to overcome this limitation and better capture the full structure of the kinetic energy cascade.

It is noteworthy that the energy injection scales diagnosed using third-order structure functions (Xie & Bühler, 2019) closely match the wavelengths of maximum growth from the linear stability analysis. While this agreement may suggest a strong

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connection between baroclinic instability and energy input, it does not imply that the system has reached a quasi-equilibrium state in the sense of idealized QG turbulence.

In fully equilibrated, forced–dissipative QG systems (e.g., Larichev & Held, 1995), energy is continuously injected at the scale of the most unstable baroclinic modes, but the system evolves toward a quasi-stationary state in which an inverse cascade consolidates energy at larger scales. In contrast, in the Iceland Basin and Irminger Sea, the injection is not steady: seasonally varying and spatially inhomogeneous forcing repeatedly excites baroclinic growth at different depths and scales. This variability prevents the system from establishing a single long-term equilibrated energy-containing scale, and the diagnosed injection scales instead reflect persistent quasi-linear baroclinic growth.

Independently of the injection-scale diagnostics, another aspect of the energy cascade dynamics emerges from the seasonal coupling between submesoscale and mesoscale kinetic energy, especially evident in fall. In mid- and low-latitude regions, mesoscale EKE typically increases about one month after the peak in submesoscale KE, reflecting the role of inverse cascading (Dong et al., 2020). In our observations, such a lag is evident in spring, with submesoscale KE peaking in March and mesoscale EKE in April. However, this pattern is absent in fall. During September to November, submesoscale energy injections intensify, but mesoscale EKE remains persistently weak. This asymmetry likely reflects a seasonal shift in the dominant mesoscale energy source. In the Iceland Basin, barotropic instability contributes approximately 50% of mesoscale eddy energy (Zhao et al., 2018), contrasting with baroclinically dominated regions elsewhere (Ferrari & Wunsch, 2009). The fall weakening of the North Atlantic Current (NAC), which modulates barotropic instability, may also suppress mesoscale EKE growth (Yaremchuk et al., 2001). Given that our structure-function-based diagnostics primarily capture baroclinic energy conversion, this mismatch between submesoscale and mesoscale KE variations in fall likely reflects a seasonal shift in the dominant instability pathway. Although the springtime lag between submesoscale and

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mesoscale KE suggests the action of an inverse cascade, as supported by prior model studies (Dong et al., 2020; Schubert et al., 2023), the current dataset lacks the temporal resolution required to estimate a cascade timescale based on energy fluxes. As such, a direct comparison between the observed lag and that inferred from flux diagnostics remains beyond the scope of this study.

Given that our structure-function-based diagnostics identify energy injection at scales consistent with those predicted for baroclinic instability, the observed mismatch between submesoscale and mesoscale KE variations in fall likely reflects a seasonal shift in the dominant instability pathway. Although the springtime lag between submesoscale and mesoscale KE suggests the action of an inverse cascade, as supported by prior model studies (Dong et al., 2020; Schubert et al., 2023), we emphasize that baroclinic instability is not the sole pathway of energy injection. Rather, the correspondence between diagnosed injection scales and theoretical instability scales is suggestive of baroclinic growth, but other processes may also contribute. Furthermore, the current dataset lacks the temporal resolution required to estimate cascade timescales from energy fluxes, and a direct comparison between observed lags and flux-derived estimates is beyond the scope of this study.

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Data availability statement.
Surface geostrophic velocity fields used in this study are available for download from the Copernicus Marine and Environment Monitoring Service (CMEMS) at the following link: https://data.marine.copernicus.eu/product/SEALEVEL_GLO_PHY_L4_MY_008_047.
The World Ocean Atlas (WOA) climatology dataset used in this study can be downloaded from: https://www.ncei.noaa.gov/access/world-ocean-atlas-2023/bin/.
The processed SADCP-based diagnostic data and codes in this study have been made publicly available at: https://github.com/jwchai/Characteristics-of-energy-cascade-in-Iceland-Basin-and-Irminger-Sea.

Appendix A. Error Propagation in Structure Function Estimation

A.1. Second-Order Structure Function

The second-order structure function used in this study is defined as:

$$D_2(r) = (\delta u^2 + \delta v^2)$$

To estimate the contribution of instrumental noise, we assume that the measured velocity increments contain uncorrelated, zero-mean Gaussian noise:

$$\delta u = \delta u_{ture} + \varepsilon u \text{ , where } \varepsilon u \sim \mathcal{N}(0, \sigma_u^2)$$

$$\delta v = \delta v_{ture} + \varepsilon v \text{ , where } \varepsilon v \sim \mathcal{N}(0, \sigma_v^2)$$

The expectation of each squared term becomes:

$$\delta u^2 = \delta u_{ture}{}^2 + \sigma_u^2$$

$$\delta v^2 = \delta v_{ture}{}^2 + \sigma_v^2$$

Thus, the total bias introduced by noise is:

$$\Delta D_2 = \sigma_u^2 + \sigma_v^2$$

Assuming $\sigma_u = \sigma_v = 0.004 m/s$. For error estimation, we assume a representative sample size of 1000 velocity difference pairs per bin, which is consistent with the winter minimum. In other seasons (spring, summer, and fall), the number of samples typically exceeds 2000 in each spatial bin.

The standard error becomes：

$$SE(D_2) \approx 1 \times 10^{-6} m^2/s^2$$

This is significantly smaller than typical structure function values (~10^{-2} m^2/s^2), and does not affect scaling behavior.

A.2. Third-Order Structure Function

The third-order structure function is defined as:

$$D3(r) = \delta u(\delta u^2(r) + \delta v^2(r))$$

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To estimate the propagated error, we again assume independent Gaussian noise:

$$\delta u = \delta u_{ture} + \varepsilon u \text{ , where } \varepsilon u \sim \mathcal{N}(0, \sigma_u^2)$$

$$\delta v = \delta v_{ture} + \varepsilon v \text{ , where } \varepsilon v \sim \mathcal{N}(0, \sigma_v^2)$$

Let:

$$f(\delta u, \delta v) = \delta u(\delta u^2(r) + \delta v^2(r))$$

Using first-order Taylor expansion, the variance of $f(\delta u, \delta v) = \delta u(\delta u^2$ is approximated as:

$$Var(f) \approx (\partial f/\partial \delta u)^2 \sigma_u^2 + (\partial f/\partial \delta v)^2 \sigma_v^2$$

Compute partial derivatives:

$$\partial f/\partial \delta u = 3\delta u^2 + \delta v^2$$

$$\partial f/\partial \delta v = 2\delta u \ \delta v$$

Then:

$$Var(D_3) \approx (3\delta u^2 + \delta v^2)\sigma_u^2 + 2\delta u \delta v \sigma_v^2$$

Assuming representative values, $\sigma_u = \sigma_v = 0.004 m/s$, $\delta u = \delta v = 0.04 m/s$。For error estimation, we assume a representative sample size of 1000 velocity difference pairs per bin, which is consistent with the winter minimum. In other seasons (spring, summer, and fall), the number of samples typically exceeds 2000 in each spatial bin.

The standard error becomes：

$$SE(D_3) \approx \sqrt{Var(D_3)} \ / \ \sqrt{N} \approx 3 \times 10^{-7} m^3/s^3$$

Appendix B. Isotropy Verification

To assess the validity of the isotropy assumption, we evaluate three commonly used isotropy metrics following the methodologies of Bühler et al. (2017) and Soares et al. (2022). These include the difference between along-track and across-track velocity variances, $\mathrm{E}(u^2 - v^2)$, their covariance $\mathrm{E}(uv)$, and the real and imaginary parts of their cross-spectrum, along with the squared spectral coherenceγ^2. Here u and v denote the along-track and across-track velocity, respectively. Figure B1 presents these diagnostics for the Iceland Basin and the Irminger Sea across all seasons. In both regions, the covariance term $\mathrm{E}(uv)$ remains close to zero (panels a–h), indicating minimal correlation between the two orthogonal velocity components. Moreover, the anisotropy measure $\mathrm{E}(u^2 - v^2)$ is generally small in magnitude and consistently

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satisfies the criterion that its value does not exceed 10% of the total kinetic energy $\mathrm{E}(u^2+v^2)$. Notably, $\mathrm{E}(u^2-v^2) \gg \mathrm{E}(uv)$, and the ratio $\mathrm{E}(uv)/\mathrm{E}(u^2-v^2)$remains on the order of 10^{-4}, well below that observed in tropical basins.

The cross-spectral analysis (panels i–p) further supports the isotropy assumption. The real part of the cross-spectrum fluctuates around zero at wavelengths below 100 km, and the imaginary part remains weak across all wavenumbers. The squared coherence γ^2 shows no significant peaks throughout the resolved wavenumber range (~0.01 to 0.1 cpkm), indicating a lack of scale-coherent coupling between the two components.

Overall, these results confirm that the isotropy assumption holds across the seasonal cycle in both the Iceland Basin and the Irminger Sea, validating the use of one-dimensional Helmholtz decomposition in our analysis.

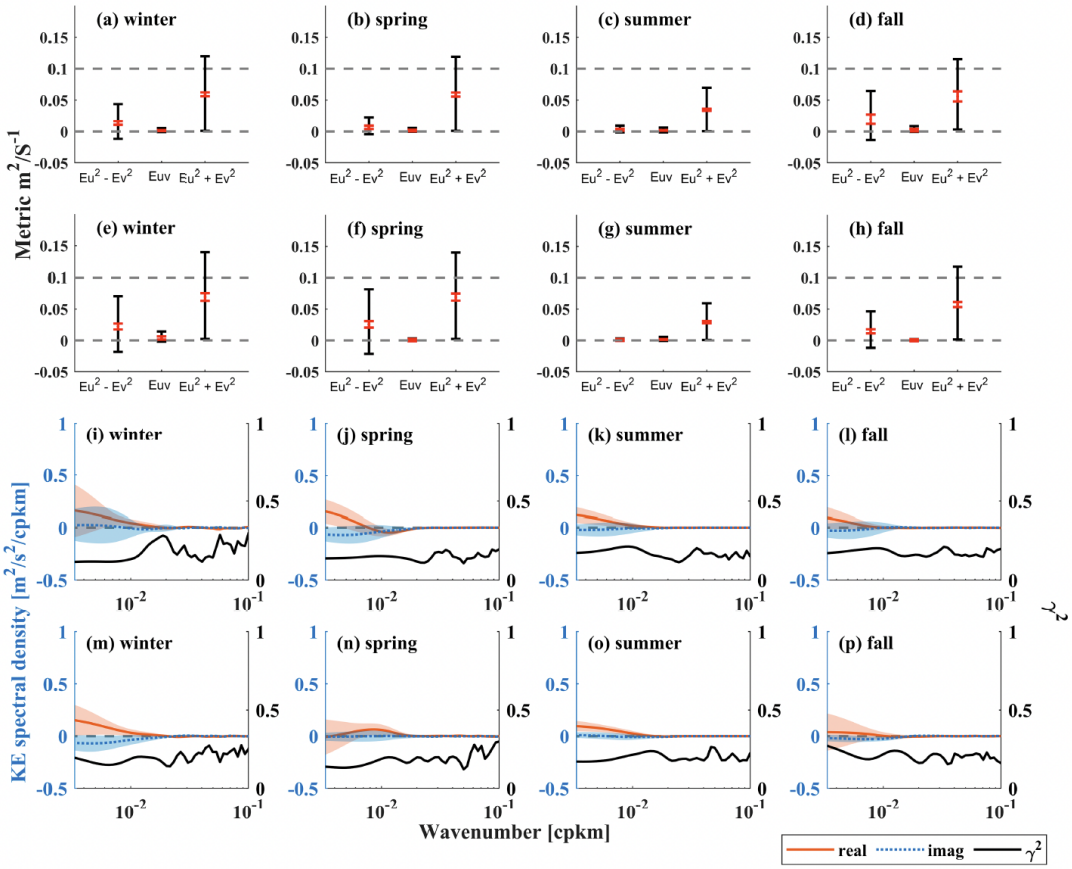

Figure B1 The metrics E($u2-v2$), E(uv), and E($u2+v2$) for different seasons in the Iceland basin(a-d) and Irminger sea(e-h). u and v here denote the along-track and across-track velocity, respectively. Red bars are twice the standard error of the mean and the black bars show the 5%–95% range. The cross-spectra (real(red) and

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imaginary(blue) components) and spectral coherence squared (γ2, right y axis) for collocated segments for different seasons in the Iceland basin(i-l) and Irminger sea(m-p). Shading gives the 95% confidence interval.

Appendix C. Validation of Structure Function Fitting

To address the reliability of the third-order structure function fitting, we provide here a representative example of the structure function D3(r) and its linear fit based on the method of Xie and Bühler (2019). Figure C1 shows D3(r)/r computed from a spring transect in the Iceland Basin and Irminger Sea. The structure function displays a clear and approximately linear trend across the relevant separation range. Despite some observational uncertainty, the fitting is stable and consistent with the assumptions required for estimating kinetic energy fluxes using structure functions. The result is comparable to the fitting quality shown in Balwada et al. (2022), supporting the validity of our method. Although our method does not resolve fine-scale fluctuations beyond the confidence intervals of the data, the successful linear fit demonstrates that meaningful kinetic energy fluxes can be reliably extracted from the observed structure functions.

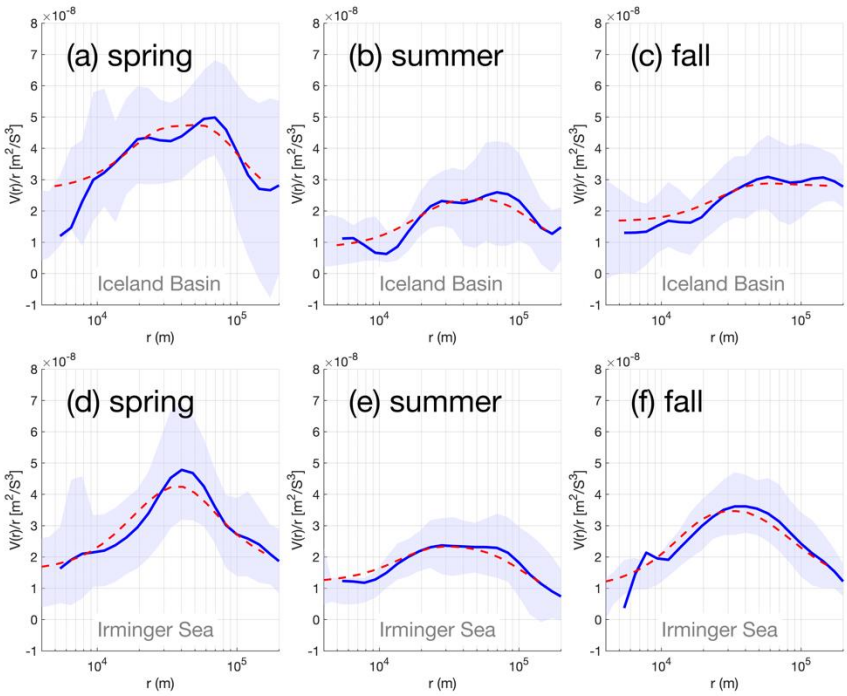

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Figure C1. Seasonal variation of the normalized third-order structure function V(r)/r in the Iceland Basin (a–c) and Irminger Sea (d–f). The blue lines represent the observed D3(r)/r, and the red dashed lines show the fitted curves based on the Xie and Bühler (2019) method. Shaded areas indicate the 95% confidence intervals estimated from ensemble averaging across transects. The close alignment between observations and fits across seasons and regions demonstrates the robustness of the fitting procedure for kinetic energy flux estimation.

Appendix D. Comparison Between Structure-Function and Coarse-Graining Methods

To evaluate the robustness of the kinetic energy (KE) cascade results, we compare two independent methods: (i) the coarse-graining framework following Srinivasan et al. (2023), and (ii) the plain least-squares fitting method described in the Supplementary Materials of Balwada et al. (2022, Section F.2), applied to the third-order structure function. Due to observational constraints, the coarse-graining method is applied in a reduced form using only zonal derivatives. Nevertheless, both approaches yield consistent results.

The coarse-graining approach diagnoses energy flux across scales in a physically consistent and Galilean-invariant manner. The total energy flux is given by:

$$P = -\left[\ \tau_{\{uu\},}u_x + \ \tau_{\{uv\},}\left(u_y + \ v_x\right) + \ \tau_{\{vv\},}v_y\right] - \ \left[\ \tau_{\{uw\},}u_z + \ \tau_{\{vw\},}v_y\right]$$

where $\tau_{\{uv\}} = \ \overline{uv} - \ \overline{u}\,\overline{v}$ denotes the subfilter stress, and the two bracketed terms represent the horizontal and vertical fluxes, respectively. Given the along-track nature of the observational data used in this study, only meridional derivatives are reliably resolved. Therefore, we compute a reduced horizontal flux component for qualitative comparison with the structure function estimates.

Figure D1 and Figure D2 present seasonal KE flux estimates in the Iceland Basin and Irminger Sea, respectively. In both regions, the structure function method (top panels) and coarse-graining method (bottom panels) exhibit similar flux structures. Across all seasons, the estimated fluxes are predominantly negative over the 1 – 100 km scale range, indicating a robust inverse cascade of kinetic energy from mesoscale to submesoscale scales. Notably, in fall, the coarse-graining method exhibits weak positive fluxes at scales larger than ~20 km. This localized deviation may arise from

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limitations in the reduced flux formulation, which only incorporates meridional derivatives. Nevertheless, given the overall agreement in flux structure and seasonal evolution between the two methods, this discrepancy is not sufficient to undermine the main conclusion regarding the forward nature of the energy cascade in both regions.

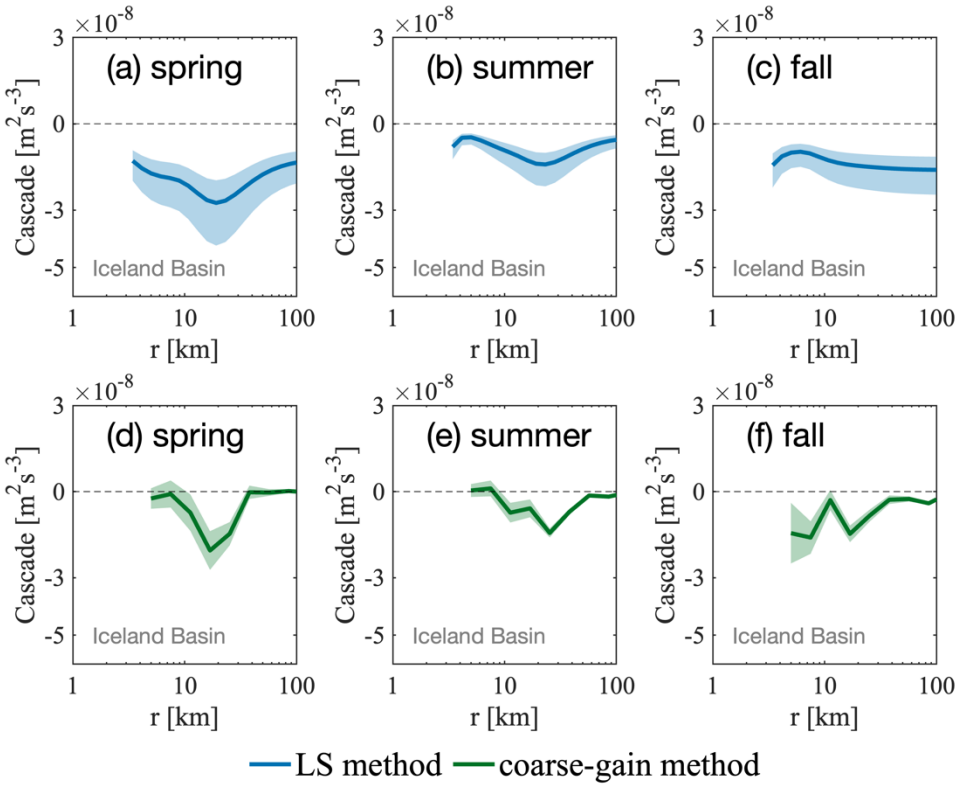

Figure D1 (a–c) show the kinetic energy cascade in the Iceland Basin during spring (a), summer (b), and fall (c), estimated using the third-order structure function least-squares method. (d–f) present corresponding seasonal results obtained using the coarse-graining approach. These two methods are compared to analyze the seasonal variations and spatial characteristics of the mesoscale-to-submesoscale energy cascade.

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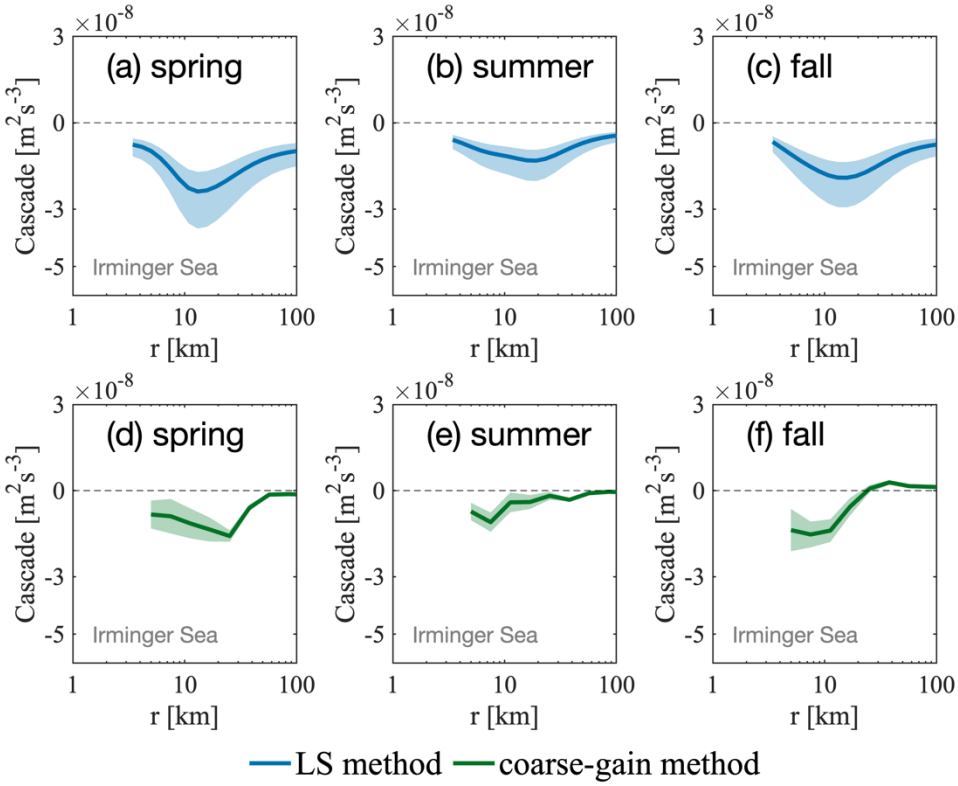

Figure D2. Same as Figure D1, but for the Irminger Sea.

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